Structured Query Language (SQL)

Essential Commands & Query Patterns

Database Systems Course

CS Department

March 17, 2026

Lecture Roadmap

1. Introduction
2. Basic Queries
3. Filtering Data
4. Aggregation
5. Joins
6. Subqueries

What is SQL?

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Today's Focus

We concentrate on **querying** and **manipulating** existing data using DML commands.

Sample Database Schema

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- `employees` (id, name, dept_id, salary, hire_date)
- `departments` (id, name, location)
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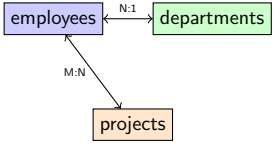

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Comparison Operators

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Use parentheses to control operator precedence!

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Limiting Results

Different SQL dialects:

```sql
-- MySQL/PostgreSQL
SELECT * FROM employees LIMIT 10 OFFSET 20;

-- SQL Server
SELECT TOP 10 * FROM employees;

-- Oracle (12c+)
```

Aggregate Functions

Common Functions

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Note: Aggregate functions ignore `NULL` values except `COUNT(*)`.

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Department Statistics

```sql
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The Golden Rule

Any column in `SELECT` that isn't an aggregate must appear in `GROUP BY` clause!

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Example

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FROM employees
WHERE hire_date > '2020-01-01'
GROUP BY dept_id
HAVING AVG(salary) > 60000;
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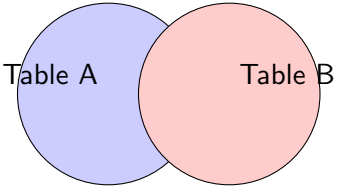

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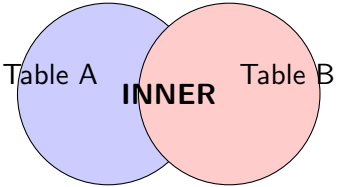

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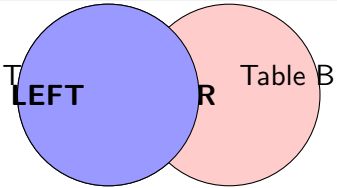

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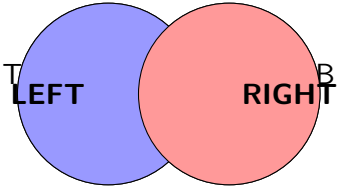

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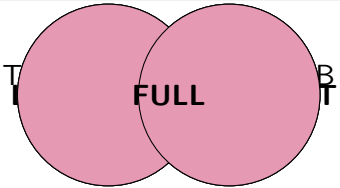

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INSERT

```sql
-- Single row
INSERT INTO employees (id, name, salary)
VALUES (101, 'Alice Smith', 75000);

-- Multiple rows
INSERT INTO employees VALUES
    (102, 'Bob', 2, 60000, '2024-01-15'),
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UPDATE & DELETE

```sql
UPDATE employees
SET salary = salary * 1.05
```

Best Practices & Tips

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- Use `EXPLAIN` to analyze queries
- Index columns in `WHERE`, `JOIN`, `ORDER BY`
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Quick Reference

`SELECT` → `FROM` → `WHERE` → `GROUP BY` → `HAVING` → `ORDER BY` → `LIMIT`

Questions?

Try These:

- Find employees earning above their department average
- List departments with no employees
- Update salaries based on tenure